

Assignment -1

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Subject - M & S

Q1. A coin is tossed twice. A random variable X represents the number of heads turning up find the discrete probability distribution for X also find mean & variance & S.D.

Soln:- $S = \{HH, HT, TH, TT\}$

X represent the no. of Heads turning up

X	0	1	2
$P(X)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

$P(X_i) \geq 0$ for all x , where $X(S)$ is range of X
 $\sum P(X_i) = 1$

$$\text{Mean } (\mu) = \sum x_i P(X_i) = 1$$

$$\text{Variance } (V) = \sum (x_i - \mu)^2 \cdot P(X_i)$$

Or

$$\sum x_i^2 \cdot P(X_i) - \left[\sum x_i P(X_i) \right]^2 = \frac{1}{2}$$

$$\text{S.D. } (\sigma) = \sqrt{V} = 0.7071$$

Q2. From a sealed box containing a dozen apples it was found that 3 apples are perished. Obtain the probability distribution of the no. of perished apple when 2 apples are drawn at random. Also find the mean & variance of this distribution

$$\text{Sol}^n:- P(x=0) = \frac{{}^3C_0 \times {}^9C_2}{{}^{12}C_2} = \frac{6}{11}$$

$$P(x=1) = \frac{{}^3C_1 \times {}^9C_1}{{}^{12}C_2} = \frac{9}{22}$$

$$P(x=2) = \frac{{}^3C_2 \times {}^9C_0}{{}^{12}C_2} = \frac{1}{22}$$

X	0	1	2
P(x)	6/11	9/22	1/22

$$\text{Mean}(\mu) = \sum x_i p(x_i)$$

$$= 0 \times \frac{6}{11} + 1 \times \frac{9}{22} + 2 \times \frac{1}{22}$$

$$= \frac{9+2}{22} = \frac{11}{22} = \frac{1}{2}, \quad E(X) = \frac{1}{2}$$

$$\text{Variance}(V) = \sum (x_i - \mu)^2 p(x_i)$$

Or

$$= E(x^2) - [E(x)]^2$$

$$= E(x^2) = \sum x^2 p(x)$$

$$= 0^2 \times \frac{6}{11} + 1^2 \times \frac{9}{22} + 2^2 \times \frac{1}{22} = \frac{13}{22}$$

$$\text{Variance} = E(x^2) - [E(x)]^2$$

$$= \frac{13}{22} - \left(\frac{1}{2}\right)^2$$

$$= \frac{15}{44}$$

Q3. If the probability of bad reaction from certain injection is 0.001, determine the chance that out of 2000 individuals, more than two will get a bad reaction.

Soln:-

$$p = 0.001, \quad n = 2000$$

$$m = np = 2000 \times 0.001$$

$$\therefore 2$$

$$P(X > 2) = 1 - P(X \leq 2)$$

$$= 1 - P(X = 0, 1, 2)$$

$\therefore$

$$\therefore P(X) = \frac{e^{-m} \lambda^x}{x!}$$

$$P(X=0) = \frac{2^0 e^{-2}}{0!} = \frac{1}{e^2}$$

$$P(X=1) = \frac{2^1 e^{-2}}{1!} = \frac{2}{e^2}$$

$$P(X=2) = \frac{2^2 e^{-2}}{2!} = \frac{2}{e^2}$$

$$P(X > 2) = 1 - \frac{5}{e^2}$$

$$= 1 - \frac{5}{7.389}$$

$$P(X > 2) = 1 - 0.68 = 0.32$$

Qf. A shop has 4 diesel generator sets which it hires every day. The demand for a generator set on an average is a poisson variate with value $5/2$. Obtain the probability that on a particular day
 (a) there was no demand
 (b) a demand had to be refused.

Soln:-

$$\text{Mean } (\mu) = \frac{5}{2} = 2.5, \text{ Capacity } k = 4$$

(a) No Demand ($X=0$)

$$P(X=0) = \frac{e^{-2.5} \times (2.5)^0}{0!} = e^{-2.5} \approx 0.0821$$

(b) ($X > 4$)

$$P(X > 4) = 1 - P(X \leq 4)$$

$$X = 0, 1, 2, 3, 4$$

$$P(X=0) = e^{-2.5}, \quad P(X=1) = e^{-2.5} \times 2.5, \quad P(X=2) = e^{-2.5} \times 3.125$$

$$P(X=3) = e^{-2.5} \times 2.607, \quad P(X=4) = e^{-2.5} \times 1.627$$

$$= 0.20521 + 0.25651 + 0.21376 + 0.13360$$

$$\approx 0.8912$$

$$P(X > 4) = 1 - 0.8912 = \underline{0.1088}$$

Q5. An AI Chatbot receives an average of 4 queries / min
Find the probability that:

(a) exactly 6 queries arrive

(b) No queries arrive in a minute

Soln:-

$$(a) P(X=x) = \frac{e^{-m} m^x}{x!}$$

$$P(X=6) = \frac{e^{-4} 4^6}{6!} = \frac{0.0183 \times 4096}{720} = \frac{74.9568}{720}$$

$$\approx 0.1041$$

$$(b) P(X=0) = \frac{e^{-4} 4^0}{0!} = 0.0183$$

Q6. If 2% of electric bulbs manufactured by certain company are defective. Find the probability that in a sample of 200 bulbs.

(a) less than 2 bulbs

(b) more than 3 bulbs are defective

Soln:-

$$p = 2\% = \frac{2}{100} = 0.02$$

$$n = 200$$

$$m = n \times p$$

$$= 200 \times 0.02$$

$$= 4$$

$$(a) P(X < 2) = P(X=0, 1)$$

$$= P(X=0) + P(X=1)$$

$$P(X=0) = e^{-4}, \quad P(X=1) = 4e^{-4}$$

$$\begin{aligned} P(X \leq 2) &= e^{-4} + 4e^{-4} \\ &= 5e^{-4} \\ &= 5(0.0183) \\ &= 0.0915 \end{aligned}$$

$$\begin{aligned} (b) \quad P(X > 3) &= 1 - P(X \leq 3) \\ &= 1 - \{ P(X=0) + P(X=1) + P(X=2) + P(X=3) \} \\ &= 1 - e^{-4} \left\{ 1 + 4 + \frac{4^2}{2!} + \frac{4^3}{3!} \right\} \\ &= 1 - 0.0183 \times (1 + 4 + 8 + 10.67) \\ &= 1 - 0.4331 = 0.5669 \approx 0.567 \end{aligned}$$

Q7. Neural network weight are uniformly distributed b/w $-2 \leq 2$
Find the probability that are randomly selected weight:

(a) Lie between $-1 \leq 1$

(b) Is greater than 1

Solⁿ:- $a = -2, b = 2$ width $= b - a = 4$

(a) b/w $-1 \leq 1$

Uniform probability $= P(c \leq X \leq d) = \frac{d-c}{b-a}$

$$P(-1 \leq X \leq 1) = \frac{1 - (-1)}{4} = \frac{2}{4} = 0.5$$

(b) Greater than 1 $\rightarrow$ Range $\rightarrow$ 1 to 2

$$P(1 \leq x \leq 2) = \frac{2-1}{4} = 0.25$$

Q8. If x is an exponential variate with mean 5, evaluate

(a) $P(0 < x < 1)$

(b) $P(-\infty < x < 10)$

(c) $P(x \leq 0 \text{ or } x \geq 1)$

Soln:- ~~$P \cdot f(x) = \int$~~

$$P(a < x < b) = \int_a^b \alpha e^{-\alpha x} \cdot dx$$

$$\mu (\text{mean}) = 5$$

$$\alpha = \frac{1}{\mu} = \frac{1}{5} = 0.2$$

$$\begin{aligned} & \int_0^1 0.2 x e^{-0.2x} \cdot dx \\ & 0.2 \int_0^1 e^{-0.2x} \cdot dx \\ & = 0.2 \left[-\frac{1}{0.2} x e^{-0.2x} \right]_0^1 = 0.2 \left[5 e^{-0.2x} \right]_0^1 \\ & = \left[-e^{-0.2x} \right]_0^1 \\ & = -e^{-0.2} - (-1) \\ & = 1 - e^{-0.2} \\ & = 1 - 0.818 \\ & = 0.181 \end{aligned}$$

$$\begin{aligned} & \int_0^1 \frac{1}{5} e^{-\frac{x}{5}} \cdot dx \\ & \frac{1}{5} \left[\frac{e^{-\frac{x}{5}}}{-\frac{1}{5}} \right]_0^1 \\ & = - \left[e^{-\frac{1}{5}} - 1 \right] \\ & = 1 - e^{-\frac{1}{5}} \\ & = 1 - 0.818 \\ & = 0.181 \end{aligned}$$

$$\begin{aligned}
 (b) \quad & \int_0^{10} \frac{1}{5} e^{-x/5} \cdot dx \\
 & \frac{1}{5} \left[\frac{e^{-x/5}}{-1/5} \right]_0^{10} \\
 & = - \left[e^{-10/5} - e^0 \right] \\
 & = 1 - e^{-2} = 0.8642
 \end{aligned}$$

$$\begin{aligned}
 (c) \quad & \int_0 + \int_1^{\infty} \frac{1}{5} e^{-x/5} \cdot dx \\
 & = e^{-0.2(1)} = 0.8187
 \end{aligned}$$

Q9. The length of the telephone ~~conversations~~ ~~conv~~ conversation in a booth has been an exponential distribution & found on an average to be 5 minutes. Find the probability that a random call made from this booth

(a) ends less than 5 min.

(b) b/w 5 & 10 min

Solⁿ:-

$$\begin{aligned}
 \text{mean} &= 5 \\
 \alpha &= \frac{1}{5}
 \end{aligned}$$

(a)

$$\int_0^5 \frac{1}{5} e^{-x/5} \cdot dx$$

$$= \frac{1}{5} \left[\frac{e^{-x/5}}{-1/5} \right]_0^5$$

$$= -[e^{-5/5} - e^0]$$

$$= -[e^{-1} - 1]$$

$$= 1 - e^{-1} = 1 - 0.367$$

$$= 0.633$$

b)

$$\int_5^{10} \frac{1}{5} e^{-x/5} \cdot dx$$

$$= \frac{1}{5} \left[\frac{e^{-x/5}}{-1/5} \right]_5^{10}$$

$$= -[e^{-10/5} - e^{-5/5}]$$

$$= [-e^{-2} - e^{-1}]$$

$$= e^{-1} - e^{-2}$$

$$= 0.232$$

Q10 A random variable X has the following probability $P(X)$ for various values of X .

X	0	1	2	3	4	5	6	7
$P(X)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$

(a) Find k

(b) Mean, Variance & S.D

(c) Evaluate $P(X < 6)$, $P(X \geq 6)$, $P(3 < X \leq 6)$

~~Ans~~

Soln :-

$$\sum P(X) = 1$$

$$(a) \quad 0 + k + 2k + 2k + 3k + k^2 + 2k^2 + (7k^2 + k) = 1$$

$$10k^2 + 9k = 1$$

$$10k^2 + 9k - 1 = 0$$

$$k = -1, \frac{1}{10}$$

$$\boxed{k = \frac{1}{10}}$$

(a) X	0	1	2	3	4	5	6	7
$P(X)$	0	0.1	0.2	0.2	0.3	0.01	0.02	0.17

$$(b) \quad \text{Mean}(\mu) = \sum x_i p(x_i) = 3.66$$

$$\text{Variance} (V) = \sum (x_i - \mu)^2 p(x_i)$$

$$= 3.4$$

$$\text{S.D} (\sigma) = \sqrt{V} = 1.8$$

$$(c) P(X < 6)$$

$$= P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4) + P(X=5)$$

$$= 0 + 0.1 + 0.2 + 0.2 + 0.3 + 0.01 = 0.81$$

$$P(X \geq 6)$$

$$= P(X=6) + P(X=7)$$

$$= 0.02 + 0.17$$

$$= 0.19$$

$$P(3 < X \leq 6)$$

$$= P(X=4) + P(X=5) + P(X=6)$$

$$= 0.3 + 0.01 + 0.02$$

$$= 0.33$$

Q11. 20 wrist watches in a box 100 are defective. If 10 watches are selected at random. Find the probability that

(a) 10 are defective

(b) 10 are good

(c) ~~at~~ at least one watch is defective

(d) almost 3 are defective

Soln :-

$$p = \frac{20}{100} = \frac{1}{5}$$

$$q = 1 - p = \frac{4}{5}$$

$$P(X=x) = {}^nC_x p^x q^{n-x} \quad x=0,1$$

$$= {}^{10}C_x \left(\frac{1}{5}\right)^x \left(\frac{4}{5}\right)^{10-x}$$

$$x = 0, 1, 2, \dots, 10$$

$$(a) \quad P(X=10) = {}^{10}C_{10} \left(\frac{1}{5}\right)^{10} \left(\frac{4}{5}\right)^0 = 1 \cdot \frac{1}{5^{10}} \times 1 = \frac{1}{5^{10}}$$

$$(b) \quad P(X=0) = {}^{10}C_0 \left(\frac{1}{5}\right)^0 \left(\frac{4}{5}\right)^{10} = \left(\frac{4}{5}\right)^{10}$$

$$(c) \quad \begin{aligned} P(X \geq 1) &= 1 - P(X < 1) \\ &= 1 - P(X=0) \\ &= 1 - {}^{10}C_0 \left(\frac{1}{5}\right)^0 \left(\frac{4}{5}\right)^{10} \\ &= 1 - \left(\frac{4}{5}\right)^{10} \end{aligned}$$

$$\begin{aligned} (d) \quad P(X \leq 3) &= P(X=0) + P(X=1) + P(X=2) + P(X=3) \\ &= {}^{10}C_0 \left(\frac{1}{5}\right)^0 \left(\frac{4}{5}\right)^{10} + {}^{10}C_1 \left(\frac{1}{5}\right)^1 \left(\frac{4}{5}\right)^9 + {}^{10}C_2 \left(\frac{1}{5}\right)^2 \left(\frac{4}{5}\right)^8 + \\ &\quad {}^{10}C_3 \left(\frac{1}{5}\right)^3 \left(\frac{4}{5}\right)^7 \\ &= 1(0.107) + 10(0.026) + 45(0.0062) + 120(0.0016) \\ &= 0.859 \text{ (approx)} \end{aligned}$$

12. If the mean & standard deviation of the no. of correctly answered questions in a test given to 4096 students are 2.5 & $\sqrt{1.85}$, Find an estimate of the no. of candidates answering correctly.

(a) 8 or more questions

(b) 2 or less.

(c) 5 questions

Soln:- $np = 2.5$
 $npq = 1.85$

$$\frac{npq}{np} = q \quad \Rightarrow \quad q = 0.75$$

$$p = 1 - q = 0.25$$

$$n = 10$$

$$(9) \quad P(X \geq 8) = \sum_{x=8}^{10} {}^{10}C_x (0.25)^x (0.75)^{10-x}$$

$${}^{10}C_8 (0.25)^8 (0.75)^2 = 45 \times 0.00001526 \times 0.5625$$

$$\approx 0.000386$$

$${}^{10}C_9 (0.25)^9 (0.75) \approx 0.0000286$$

$${}^{10}C_{10} (0.25)^{10} \approx 0.0000016$$

$$P(X \geq 8) \approx 0.000416$$

$$4096 \times 0.000416 \approx 2 \text{ Students}$$

$$(b) P(X \leq 2) = \sum_{x=0}^2 {}^{10}C_x (0.25)^x (0.75)^{10-x}$$

$${}^{10}C_0 (0.25)^0 (0.75)^{10} \approx 0.0563$$

$${}^{10}C_1 (0.25)^1 (0.75)^9 \approx 0.1878$$

$${}^{10}C_2 (0.25)^2 (0.75)^8 \approx 0.2815$$

$$P(X \leq 2) \approx 0.5256$$

$$4096 \times 0.5256 \approx 2152$$

$$(c) P(X=5)$$

$${}^{10}C_5 (0.25)^5 (0.75)^5$$

$$\approx 239 \text{ students}$$

Q13. ~~In~~ In a test on 2000 electric bulbs, it was found that life of a particular make, was normally distributed with an average life of 2040 hours & S.D of 60 hours, estimate the no. of bulbs likely to burn for

(a) more than 2150 hours

(b) less than 1950 hours

(c) more than 1920 hours & but less than 2160 hours

Solⁿ:- $(\mu) \text{ mean} = 2040$

$$SD (\sigma) = 60$$

$$Z = \frac{X - \mu}{\sigma}$$

a) $(x > 2150)$

$$z = \frac{2150 - 2040}{60} = \frac{110}{60} = 1.833$$

$$= 0.5 - P(z < 1.833)$$

$$= 0.5 - 0.4669 = 0.0331$$

The no. of bulbs expected to burn for more than

$$2150 \text{ hours} = 0.0331 \times 2000 \\ = 66.2 \text{ (approx)}$$

b) $(x < 1950)$

$$z = \frac{1950 - 2040}{60} = -\frac{90}{60} = -1.5$$

The curve is symmetric, so Area to -1.5 is the same as area to $+1.5$, find $+1.5$

Area from 0 to 1.5 is 0.4332

We want less than -1.5 which is the left.

$$z = \frac{x - \mu}{\sigma} = -1.33$$

$$P(x < 1950) = P(z < -1.33) = P(z > 1.33)$$

$$= 0.5 - P(0 < z < 1.33)$$

$$= 0.5 - 0.4082 \text{ (table value to } z = 1.33)$$

$$= 0.0918$$

the no. of bulbs expected to burn less than 1950 hours

$$= 0.0918 \times 2000 = 183.6 \text{ (approx)}$$

(c) When $X = 1920$

$$Z = \frac{X - \mu}{\sigma} = -2$$

When $X = 2160$, $Z = 2$

$$P(1920 < X < 2160) = P(-2 < Z < 2) = 2 \times P(0 < Z < 2)$$

$$= 2 \times 0.4772$$

$$= 0.9544$$

~~The no.~~ The required no. of bulbs = 0.9544×2000
 $= 1909$ nearly.

Q14 If X is a normal ~~variable~~ variate with mean 30 & standard deviation 5, then find the probabilities that.

(a) $26 \leq X \leq 40$

(b) $X \geq 45$

(c) $|X - 30| > 5$

Solⁿ:- $\mu = 30$, $\sigma = 5$

(a) $(26 \leq X \leq 40)$

$$Z = \frac{26 - 30}{5} = -0.8, \quad Z = \frac{40 - 30}{5} = 2$$

$$P(-0.8 < Z < 2) = \text{Area}(0 \rightarrow 0.8) + \text{Area}(0 \rightarrow 2)$$

$$A(0.8) = 0.2881$$

$$A(2.0) = 0.4772$$

$$P = 0.2881 + 0.4772 = 0.7653$$

$$1 \quad X \geq 45$$

$$Z = \frac{45 - 30}{5} = 3.0$$

$$P(Z > 3.0) = 0.5 - \text{Area}(0 \rightarrow 3.0)$$

$$A(3.0) = 0.4987$$

$$p = 0.5 - 0.4987 = 0.0013$$

(C' $|X - 30| > 5$ (This implies deviation from mean is > 5)

X is above the mean by more than 5 : $X > 30 + 5 \Rightarrow X > 35$

X is below the mean by less than 5 : $X < 30 - 5 \Rightarrow X < 25$

For $X = 35$

$$Z = \frac{35 - 30}{5} = \frac{5}{5} = 1.0$$

for $X = 25$

$$\frac{25 - 30}{5} = \frac{-5}{5} = -1$$

— We need the area in tail : from right of 1.0 & area to the left of -1.0

— The curve is ~~symmetric~~ $A(\text{Right}) = A(\text{left})$

— We can calculate one tail & multiply by 2.

$$A(0 \rightarrow 1.0) = 0.3413$$

Right tail Area = Half curve - (Center of z)

$$\text{Tail Area} = 0.5 - 0.3413 = 0.1587$$

Total probability:

$$P = 2 \times 0.1587 = 0.3174$$

Q15 The weight of the ball bearings produced in a certain factory are normally distributed with mean 0.5 N & S.D 0.002 N . Find the % of ball bearings produced & weights

(a) less than 0.498 N

(b) more than 0.501 N

(c) B/w 0.4995 & 0.5005 N

Soln:

$$\mu = 0.5$$

$$\sigma = 0.002$$

$$(a) \quad Z = \frac{0.498 - 0.5}{0.002} = \frac{-0.002}{0.002} = -1$$

$$P(Z < -1) = 0.5 - \text{Area}(0 \rightarrow 1.0)$$

$$A(1.0) = 0.3413$$

$$P = 0.5 - 0.3413 = 0.1587$$

Result: 15.87%

(b) More than 0.501

$$Z = \frac{0.501 - 0.5}{0.002} = \frac{0.001}{0.002} = 0.5$$

$$A(0.5) = 0.1915$$

$$P(Z > 0.5) = 0.5 - A(0 \rightarrow 0.5)$$

$$P = 0.5 - 0.1915 = 0.3085$$

Result = 30.85%

(C) b/w 0.4995 & 0.5005

$$Z_1 = \frac{0.4995 - 0.5}{0.002} = -0.25$$

$$Z_2 = \frac{0.5005 - 0.5}{0.002} = 0.25$$

$$P(-0.25 < Z < 0.25) = 2 \times A(0.25)$$

$$A(0.25) = 0.0987$$

$$P = 2 \times 0.0987 = 0.1974$$

Result = 19.74%

Shweta
13/2/26